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Stabilization of Phase Transition Process and Lattice Oxygen of O3-Layered Oxide Cathode for Sodium-Ion Battery via Dual-Doping Strategy

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Satisfying stable cycling under high voltage and efficient ion extraction/insertion kinetics are important factors in constructing high-energy density sodium-ion batteries. O3-layered oxide cathodes draw significant attention due to their high specific capacity and mature synthesis processes. However, these materials are plagued by stress accumulation from phase transitions, sluggish sodium ion diffusion kinetics, and irreversible lattice oxygen oxidation at high voltage. Herein, a $Na(Ni_{1/3}Fe_{1/3}Mn_{1/3})_{0.98}Al_{0.02}B_{0.02}O_2$ material is reported that addresses the inherent limitations of O3-layered oxides under high voltage through a rationally designed dual-element doping mechanism. Al^{3+} that occupies the transition metal layer, can strengthen TM—O bonds as well as inhibit interlayer sliding and phase transition. Meanwhile, B^{3+} doping at the interstitial sites between the transition metal layer and oxygen layer stabilizes lattice oxygen via robust B—O covalent interactions, thereby reducing lattice oxygen release and avoiding excessive decomposition of electrolyte catalyzed by reactive oxygen. Furthermore, this dual-element doping cathode achieves expanded Na—O interlayer spacing and an elevated c/a ratio, synergistically enhancing Na-ion diffusion kinetics. The $Na(Ni_{1/3}Fe_{1/3}Mn_{1/3})_{0.98}Al_{0.02}B_{0.02}O_2$ cathode exhibits a stable and reversible sodium storage structure, so that it significantly improves high-rate capability, and cycling performance. This bulk-interface synergistic bond-energy engineering provides a novel perspective for high-voltage cathodes in sodium ion batteries.

1. Introduction

Sodium-ion batteries (SIBs) exhibit compelling potential for grid-scale energy storage and low-speed electric mobility, leveraging inherent advantages including enhanced safety, abundant sodium reserves, and cost-effectiveness.[1,2] However, the larger radius of sodium ions (1.02 Å) and their high molar mass (23 g mol^{-1}) result in the reduced mass-/volumetric energy density in SIBs.[3] In addition, the higher standard redox potential of sodium (Na^+/Na: −2.74 V) further compromises the achievable energy density, failing to meet the desired performance standards. To enhance the competitiveness of SIBs, improving energy density at the electrode level is the most rapid and efficient approach. The cathode material plays an extremely crucial role in the reversible capacity, energy density, and cycle life of SIBs. Various types of compounds, including layered transition metal oxides, poly-anion compounds, and Prussian blue analogues, have been explored as potential candidates. Layered transition metal oxides with an O3-type structure (denoted as Na^xTMO_2, where x ≤ 1 and TM represents transition metals, such as $NaNiO_2$, $NaMnO_2$, $NaNi_{1/2}Mn_{1/2}O_2$ and $NaNi_{1/3}Fe_{1/3}Mn_{1/3}O_2$) are considered premier cathode materials for commercial SIBs. This is attributed to their high sodium content (0.9–1.0), which enables a high reversible specific capacity, combined with scalable synthesis methods.[4] Furthermore, compared with other cathode materials, achieving a higher charging cut-off voltage can induce more Na-ion desodiation from the O3-type layered oxides cathode, which is a feasible strategy to enhance energy density.[5,6]

The high-voltage platform of O3-type layered oxide cathodes is a mixed blessing for SIBs, due to it introduces the following issues. During the charging and discharging processes, multiple structural evolutions (phase transitions) are inevitable upon (de)sodiation, resulting in poor cycling stability and high-rate performance.[7] Taking the promising commercial potential cathode, O3-type $NaNi_{1/3}Fe_{1/3}Mn_{1/3}O_2$ (NFM) layered oxide, as an example, Qiao et al. reported that the phase transitions or

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coexistence among the O3, P3, and OP2 phases will occur during charging and discharging processes.[8] The volumetric change and local stress induced by the phase transitions can give rise to the formation of microcracks, especially during the significant evolution during the P3 to OP2 phase transformation process at high voltage (>4.1 V), which poses a major challenge to the battery's long cycle life. In addition, oxygen release may be triggered under high-voltage operating conditions, damaging the reversibility of O3-type layered cathode materials. Liu et al. showed the redox activity of oxygen in the NFM cathode between 4.0 and 4.3 V.[9] The exotic oxygen valence state of $O^{\alpha-}$ ($0 < \alpha < 2$) tends to be more mobile, leading to the diffusion of lattice oxygen to the active surface.[10] This further triggers its intense side reactions with the electrolyte, resulting in excessive consumption. Moreover, high voltage operation also exacerbates the irreversible transformation of NFM both at the surface and in bulk. Concurrently, the safety issues of batteries caused by the substantial decomposition of electrolyte and the resultant gas evolution represent a challenge that cannot be overlooked.[11] Finally, the complex Na-ion migration paths in the O-type structure intrinsically limit the Na-ion transport and impair the high-rate capability of O3-type layered cathodes.[12]

Recently, the majority of previously reported strategies have been devoted to addressing the aforementioned bottleneck issues, including surface engineering,[13] structural design,[14,15] and elemental doping.[16,17] Unlike the first two strategies, which are hampered by inconsistent material synthesis and complex procedures, elemental doping offers an efficient and scalable means to improve cathode performance at high-voltage. Specifically, high-valence metal (M) ion doping, such as Ti^{4+},[18,19] Sn^{4+},[20] and Nb^{5+},[21] which possess strong M—O bond energies, can inhibit the slide of TMO_6 slab by occupying positions within the TM layers. Specifically, certain ions like K^+[22] and Zn^{2+}[23] can mitigate structural degradation by occupying sites within the Na layers, thereby stabilizing the framework against collapse induced by extensive Na-ion desodiation at high voltages, acting as "pillars". Notably, in LIBs, B ions with a small ionic radius (0.23 Å) and strong bond energies (B—O 809 kJ mol^{-1}) have been reported to occupy tetrahedral interstitial sites, achieving robust stabilization of lattice oxygen. Li et al. investigated the effects of B atom doping into the Li_2RuO_3 cathode,[24] which can modify the electronic structure of the Ru—O system and lower the energy of the oxygen 2p band, thereby mitigating the irreversible oxidation of oxygen. In SIBs, Vaalma et al. reported that B doping enhanced the cycling reversibility of the P2-type $Na_{2/3}MnO_2$ cathode material.[25] However, in the case of O3-type layered oxides exemplified by NFM, the mechanism of B doping in stabilizing the lattice oxygen has yet to be explored. Moreover, merely addressing the issue of reactive oxygen species is insufficient to comprehensively enhance the performance of layered oxides under high-voltage conditions. This also reflects the limitations of conventional single-element doping, which typically yields only one singular effect.[22,26] An efficient and cost-effective strategy also needs to be identified to address the issue of structural phase transition caused by the slippage of the TMO_6 slab in layered oxide cathodes.[27] Consequently, co-doping with multiple function elements could induce synergistic effects and serve as a comprehensive solution to simultaneously address the poor high-rate capability, structural instability, and surface side reactions of high-voltage O3-type layered cathodes.

In this work, we report that trace amounts of B and Al co-doping (0.02 mol of each) that enables low-strain structural evolution and high stability interface evolution in O3-type layered NFM under 4.3 V high voltage operation, systematically addressing critical challenges of irreversible lattice oxygen oxidation, phase-transition-induced stress accumulation, and sluggish Na-ion diffusion in conventional O3-type layered cathodes. Departing from single-element doping limitations, we innovatively propose the following synergistic mechanisms: 1) The occupancy of Al^{3+} in TM layers confers the strong Al—O bonding strength of 512 kJ mol^{-1}, higher than Ni/Fe/Mn—O bonds (392, 409 and 402 kJ mol^{-1}, respectively), thereby inhibiting slippage of TMO_6 slabs, reducing the *c*-axis variation ($\Delta\theta$: 2.1° → 1.7°) during sodiation/desodiation, and alleviating biphasic strain; 2) The occupancy of B^{3+} at TM—O interstitial sites enhances lattice oxygen stability via strong B—O covalent bonding (809 kJ mol^{-1}). Consequently, the formation energy of oxygen vacancies increases from 1.7 to 2.4 eV, reducing the release of reactive oxygen species and effectively suppressing electrolyte decomposition, as evidenced by a 71.2% decrease in CO_2 evolution; 3) The optimized Na-ion diffusion channels through coordinating TMO_6 slab contraction and Na layers expansion, resulting in the enhanced Na-ion diffusivity (112.3 mAh g^{-1} at 5C rate). The experimental results show that B and Al co-doping endows the $Na(Ni_{1/3}Fe_{1/3}Mn_{1/3})_{0.98}Al_{0.02}B_{0.02}O_2$ cathode with superior overall performance in the 2–4.3 V range. The initial Coulombic efficiency (ICE) increases to 92.5% (from 82.8%), and the capacity retention after 200 cycles at 2C nearly doubles compared with NFM cathodes. Moreover, a reversible capacity of 87.6 mAh g^{-1} is maintained at 10C. Furthermore, we revealed a bulk-interface co-stabilization mechanism: Al reinforces TM—O mechanical integrity, meanwhile B passivates oxygen-loss-driven interfacial degradation, synergistically lowering the activation energy (Ea) and enabling its long-term cycle. This atomic-scale "bond-energy engineering" paradigm establishes a generalizable pathway for resolving high-voltage challenges in layered oxide cathodes and realizing high-energy-density SIBs, strategically integrating scalable synthesis methods with inherent commercial viability.

2. Discussion

In this study, a series of B or/and Al doped O3-type layered oxide materials, including $NaNi_{1/3}Fe_{1/3}Mn_{1/3}O_2$ (NFM), $Na(Ni_{1/3}Fe1_{/3}Mn_{1/3})_{0.98}Al_{0.02}O_2$ (NFMA), $NaNi_{1/3}Fe_{1/3}Mn_{1/3}B_{0.02}O_2$ (NFMB), and $Na(Ni_{1/3}Fe_{1/3}Mn_{1/3})_{0.98}Al_{0.02}B_{0.02}O_2$ (NFMAB) were synthesized via high-temperature solid-state reactions (the detailed preparation methods were provided in the Materials Synthesis section).

We employed density functional theory (DFT) calculations to investigate potential doping mechanisms of B and Al in the bulk structure. Considering the relatively large ionic radius of Al^{3+} (0.51 Å), we primarily evaluated its occupation in either Na layers or TM layers sites. Given the small ionic radius of B^{3+} (0.27 Å), both interstitial doping at Na—O or TM—O layer sites and substitutional doping at Na/TM layer positions were systematically evaluated with site-preference calculations (the computational details were provided in the Supporting Information).

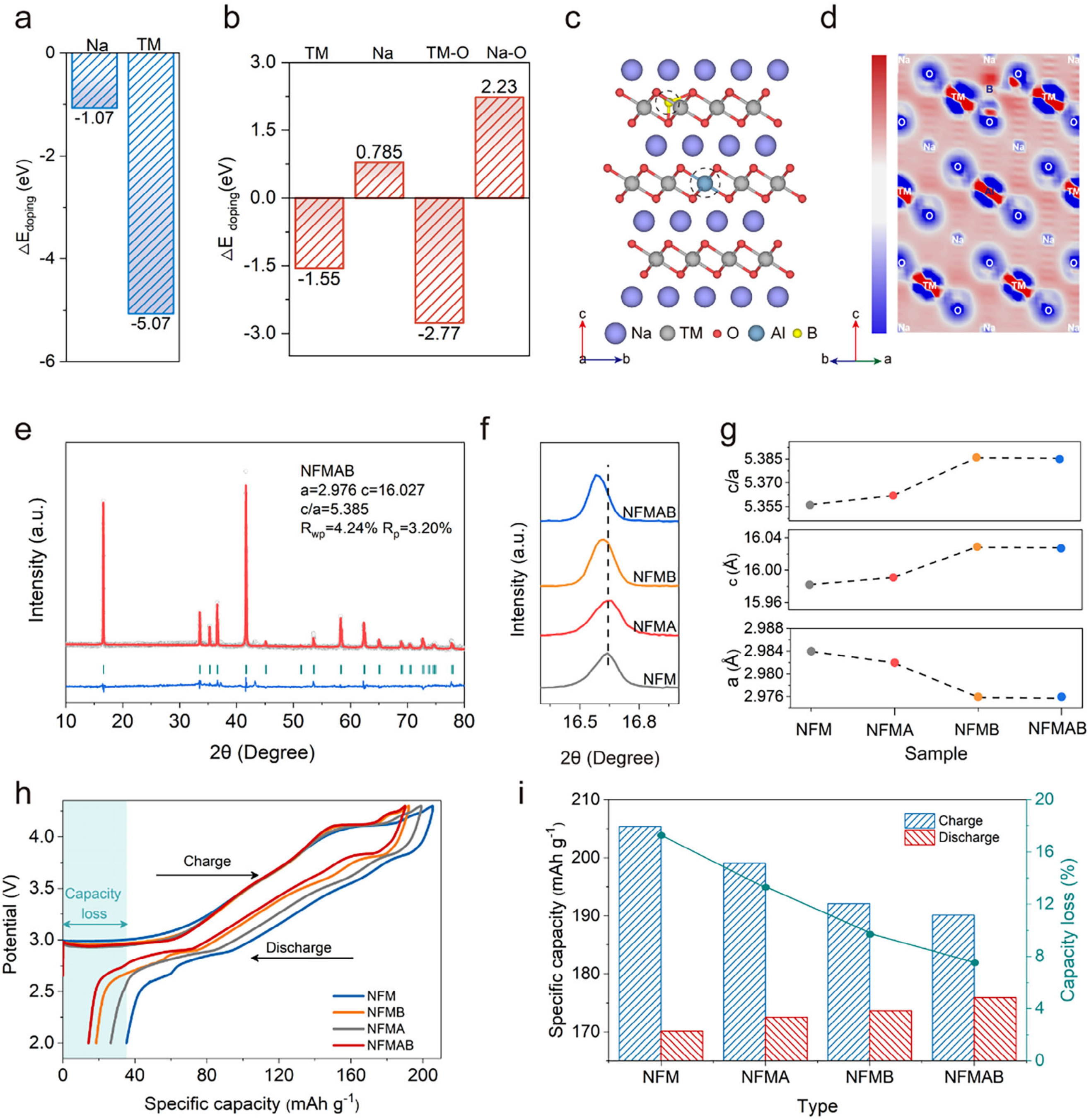

Figure 1. Occupancy energies of doping elements at different sites: a) Al and b) B; c) Schematic of elemental doping sites in NFMAB from the view of bc plane; d) Differential charge density of NFMAB from the view of abc plane; e) XRD refinement pattern of NFMAB material; f) Partial magnification of the XRD refinement patterns in cases of NFM, NFMA, NFMB, and NFMAB; g) The *a*-axis, *c*-axis, and c/a ratios of NFM, NFMA, NFMB, and NFMAB; h) Charge–discharge profiles of NFM, NFMA, NFMB and NFMAB half cells; i) Initial capacities and capacity loss for NFM, NFMA, NFMB, and NFMAB half cells.

The calculated occupancy energies of doping elements reveal distinct site preferences: Al doping exhibits occupancy energies of −1.07 and −5.07 eV for Na site and TM site respectively, indicating a strong thermodynamic preference for TM-site incorporation (**Figure 1**a). B atom exhibits the lowest occupancy energy (−2.77 eV) when it occupies the interstitial site between TM and O layers, compared with those of Na-layer substitution (0.785 eV), TM-layer substitution (−1.25 eV) and Na—O interstices (2.23 eV), confirming that the preferential B occupancy is at TM—O interstitial site (Figure 1b).

The preferred doping sites within the layered framework in the bc plane are depicted in Figure 1c, revealing the distinct spatial distribution of Al and B in TM layers and TM—O layers interstitial positions, respectively. We also provide a perspective of the doped NFMAB cathode in the ab plane in Figure S1 (Supporting Information).

The differential charge density analysis (Figure 1d) reveals that the introduced B^{3+} ions, owing to the stronger covalency of B—O bonds, which donate additional electrons to oxygen atoms through charge transfer. This mechanism significantly reinforces the oxygen ligand framework, thereby effectively mitigating oxygen loss from the lattice and enhancing structural stability. The differential charge density of the NFM sample is presented in Figure S2 (Supporting Information). To explore the influence of heterogeneous-element doping on crystal structure, Rietveld refinement was performed, yielding reliable results (R_{wp} < 10%) with crystallographic parameters summarized in Tables S1 and S2 (Supporting Information). As shown in Figure 1e and Figure S3a—c (Supporting Information), all the samples have a hexagonal O3-type α-$NaFeO_2$ structure with space group R-3m, matching PDF # 01-076-2299. Incorporation of Al^{3+} induces a slight contraction in the ab plane lattice parameter (from 2.984 Å in pristine NFM to 2.982 Å in NFMA). This structural change is due to the higher bond strength of Al—O (relative to TM—O bonds), leading to the reduced interatomic space and the subsequent contraction of TM layer in ab plane. The comparable bond energy and ionic radius of Al^{3+} with TM ions (Ni^{2+}, Fe^{3+}, Mn^{4+}) ensure good compatibility, resulting in minimal changes to the crystal structure by Al doping. As shown in the locally enlarged image of X-ray diffraction (XRD) (Figure 1f) from 16.3° to 17.0°, the introduction of B^{3+} causes a significant left shift of (003) peak, indicating the successful B doping and the expansion along *c*-axis. Combined with the coordinate changes from XRD refinement, the results show that B and Al co-doping reduces the TMO_6 octahedral plate thickness from 2.354 to 2.311 Å and increases the NaO_2 plate spacing from 2.974 to 3.039 Å. The bond lengths and interlayer spacing parameters of the refined material are provided in Table S3 (Supporting Information), which demonstrates that strong B—O bonds can contract the TM—O layer and expand the Na—O layer. As shown in Figure 1g, the co-doping increases the *c*-axis and contracts the *a*-axis. The elevated c/a ratio in the case of NFMAB, compared with that of the control samples (NFM and NFMA), can facilitate sodium-ion conduction and enhance the intrinsic kinetics of layered oxides.

Scanning electron microscopy (SEM) images revealed that the NFM and NFMAB samples were composed of particles sized between 0.8 and 1.0 µm (Figure S4a,b, Supporting Information). No significant morphological differences are observed between the NFM and NFMAB samples, indicating that the B and Al co-doping has minimal impact on their morphology. The NFMA and NFMB samples likewise display no notable changes or additional unique features. Energy dispersive spectrometer (EDS) mapping confirms the changes in chemical composition following doping and reveals that, for the co-doped samples, B and Al, as well as Na, Ni, Fe, Mn, and O, are homogeneously dispersed (Figure S4c—h, Supporting Information). The transmission electron microscopy (TEM) image in Figure S4i (Supporting Information) displays clear lattice fringes that correspond well to the characteristic (003) peak in the PDF card # 01-076-2299 of O3-type α-$NaFeO_2$, in agreement with the refined results of XRD. Fast Fourier transform analysis of lattice spacings (Figure S4j, Supporting Information) further demonstrates uniform periodicity, confirming phase homogeneity and high crystallinity in the synthesized material. As analyzed by inductively coupled plasma (ICP) (Figure S4k, Supporting Information), the actual composition ratio of the material is $Na_{1.014}Ni_{0.3203}Fe_{0.3195}Mn_{0.3204}Al_{0.0192}B_{0.0206}O_2$, which is consistent with the experimental design.

The initial galvanostatic charge/discharge curves of NFM, NFMA, NFMB, and NFMAB half cells, measured at 0.1C rate within a voltage window of 2–4.3 V (Figure 1h), reveal distinct electrochemical characteristics arising from B or/and Al doping. It can be observed that the introduction of B leads to a capacity decrease in the high-voltage region (4.1–4.3 V) during the charging process, corresponding to the suppression of irreversible oxidation of lattice oxygen and the alleviation of excessive decomposition of the electrolyte, thereby achieving a reduced capacity loss. As shown in Figure 1i, the initial reversible capacity of NFM, NFMA, NFMB, and NFMAB half cells are 170.1, 172.5, 173.6, and 175.9 mAh g^{-1}, and their irreversible capacity losses are 35.3, 26.5, 18.4, and 14.2 mAh g^{-1}, respectively. The ICE of NFMAB cathode is 92.5%, which is much higher than that of the control sample NFM (82.8%). The above results prove that the synergistic effect of B and Al co-doping achieves a dual improvement in the redox reversibility under high voltage and the compatibility with the electrolyte by stabilizing the lattice oxygen and suppressing the interfacial reaction.

To systematically evaluate the electrochemical implications of samples by B or/and Al doping, the galvanostatic charge–discharge cycling tests were performed on 2032-type coin cells by assembling with Na metal as a counter electrode. **Figure 2a** shows the cycling performance of the pristine NFM and the doped samples (NFMA, NFMB, and NFMAB) within the voltage range of 2–4.3 V, following activation at 0.1C and during long-term cycling at 2C. As mentioned above, the NFMAB sample mitigated the side reactions at high voltage, achieving a significant improvement in the ICE, thereby avoiding the capacity loss due to the irreversible lattice oxygen oxidation. Concurrently, the lattice expansion induced by B doping facilitates the conduction of Na-ion, so that the NFMAB cathode achieved a reversible capacity of 149.9 mAh g^{-1} at 2C, significantly surpassing the 108.23 mAh g^{-1} observed for the pristine NFM. The pristine NFM electrode suffers severe capacity fade during cycling, retaining merely 31.3% of its initial capacity after 200 cycles. In contrast, the B and Al co-doped NFMAB exhibit markedly improved cyclability and a high cycling retention rate of 63.1% compared with that of NFMA (59.1%) and NFMB (24.8%), respectively. The enhancement stems from the fact that the incorporation of Al-element strengthens the TM—O bonds. This strengthening effect suppresses the repeated phase transitions resulting from the sliding of the TMO_6 slab, thereby averting the decline in cycle life. In contrast, materials without Al doping exhibited varying degrees of abrupt capacity drop during long cycling.

As shown in Figure 2b, the NFMAB sample achieved a stable CE (99.7%) over 200 cycles. It is because Al doping stabilizes the crystal structure, meanwhile, B doping effectively stabilizes the lattice oxygen at high voltages, preventing issues such as gas evolution and catalytic decomposition of the electrolyte. In contrast, both the pristine sample and the samples with single-element doping encountered the issue of deteriorating Coulombic efficiency in the latter stages of cycling.

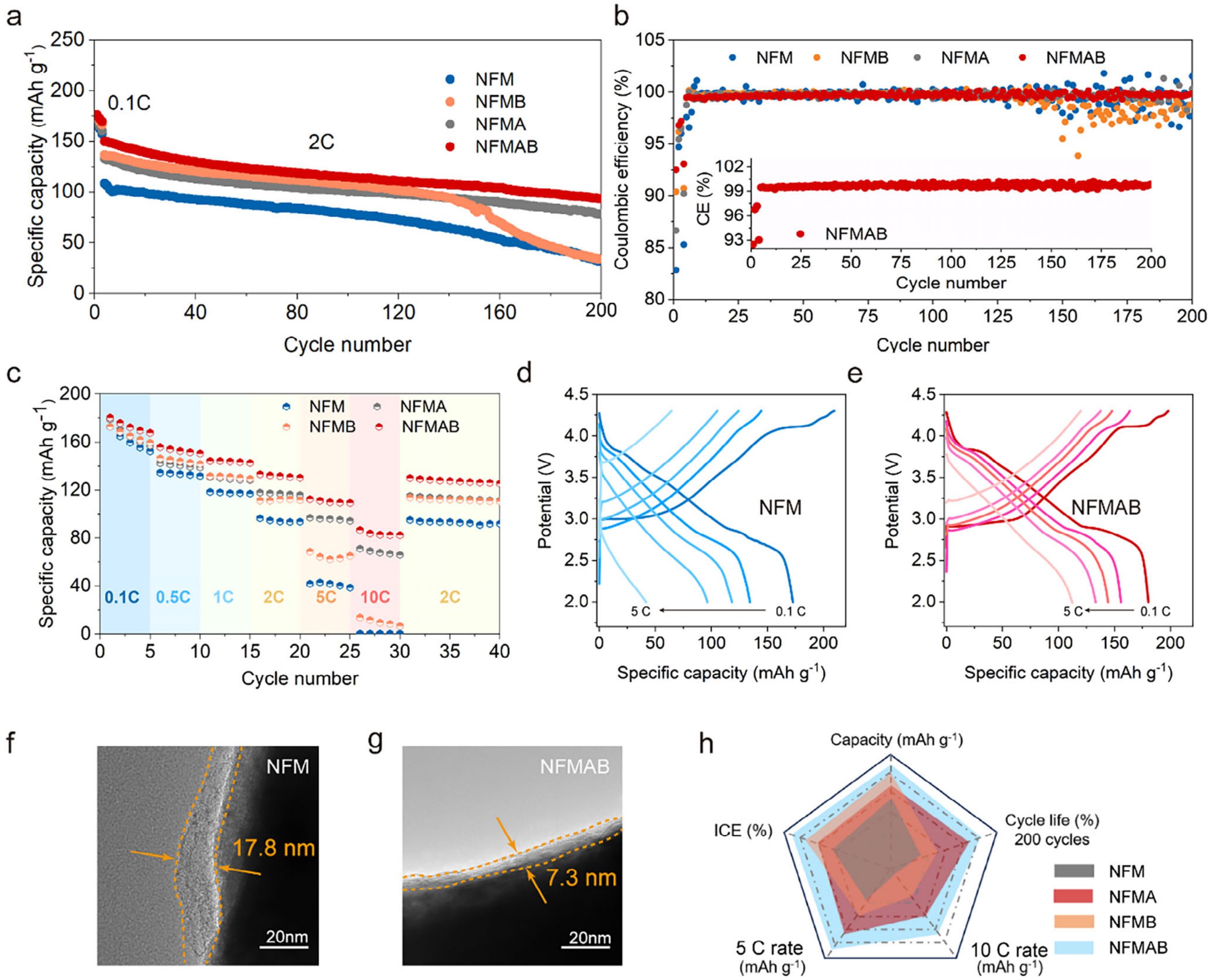

Figure 2. Electrochemical performance of NFM, NFMA, NFMB, and NFMAB half cells within a 2–4.3 V voltage range. a) Long-term cycling profiles at 2C after three activation cycles at 0.1C; b) Coulombic efficiency profiles; c) Rate capability from 0.1C to 10C; d) NFM and e) NFMAB charge/discharge curves at varying rates (0.1–5C); TEM analysis after 10 cycles for f) NFM and g) NFMAB half cells; h) Multidimensional comparative analysis of electrochemical properties for NFM, NFMA, NFMB, and NFMAB half cells.

Figure 2c illustrates the rate capabilities of NFM, NFMA, NFMB, and NFMAB cathodes at various current densities ranging from 0.1 to 10C. The specific discharge capacity of the original NFM only reaches 41.8 mA h g^{-1} at a current rate of 5C. In contrast, all the doped samples exhibit higher specific capacities. Notably, the NFMAB cathode exhibits a reversible discharge capacity 2.6 times higher than that of the pristine sample at 5C, and even maintains a high capacity of 87.6 mAh g^{-1} at 10C. As can be seen from the charge–discharge profiles at different rates (Figure 2d,e), although the reversible capacities of all half cells decrease significantly with the increase in current density, the NFMAB sample always maintains an advantage in capacity at different rates. Compared with the NFM control sample, the enhanced high-rate capability is attributed to the broadened Na-ion transport pathways and stable ion channels. This is mainly due to the enhanced rigidity of the TM layers' structure and the anchoring effect of lattice oxygen induced by B and Al co-doping. They synergistically suppress phase transition stress accumulation during cycling and maintain the durability of the fast Na-ion diffusion network. TEM shows (Figure 2f,g) that NFM electrodes form a non-uniform cathode electrolyte interphase (CEI) layer (17.8 nm), while NFMAB electrodes have a thinner (7.3 nm) and denser CEI layer, indicating that the interfacial side reactions caused by the decomposition of the electrolyte oxidized by reactive oxygen species are delayed. As shown in Figure 2h, a comprehensive comparison of the electrochemical performance of the four materials across different dimensions reveals that the B and Al co-doped cathode NFMAB material exhibits significant improvements in ICE, rate capability, and long cycling stability. These results demonstrate that dual-element doping with B and Al is an effective strategy for synergistically improving the bulk and interfacial issues of NFM materials at high voltages. We also place the battery performance comparison with previous works in Figure S5 (Supporting Information).

To further elucidate the mechanism of synergistic doping by dual elements, more in-depth electrochemical analyses are conducted. Cyclic voltammetry (CV) profiles were measured during the initial redox processes to reveal electrochemical characteristics between 2.0 and 4.3 V at a scan rate of 0.1 mV s^{-1} (vs Na^+/Na), as shown in **Figure** 3a,b. In the NFMAB half cell, an anodic peak at 3.393 V is observed during the first cycle, corresponding to the desodiation process associated with the oxidation of Ni^{2+}[8]. This peak appears at a lower voltage compared with that of the reference sample (3.471 V, Figure 3b), suggesting reduced sodium ion diffusion resistance and accelerated desodiation kinetics in the NFMAB material. Concurrently, the polarization of the battery decreases from 0.537 to 0.432 V. In 10 CV scans, the NFMAB sample exhibits superior peak overlap, indicating a stable potential difference between the oxidation and reduction peaks. This suggests that irreversible side reactions, such as electrolyte decomposition and irreversible phase transitions, are effectively suppressed. Conversely, the reference sample shows an increasing polarization during the cycling process.

Electrochemical impedance spectroscopy (EIS) measurements across cycling stages (Figure 3c,d) reveal consistently lower interfacial resistance in the NFMAB half cell compared with the pristine counterpart. This improvement likely stems from a mitigated accumulation of decomposition-derived interphase products. The high bonding energy between B and O mitigates the release of active oxygen at high voltages, thereby reducing the reactivity associated with electrolyte oxidation by active oxygen under high-voltage conditions- a critical mechanism for minimizing interfacial resistance formation.

The variable-scan-rate CVs of NFM and NFMAB cathodes are measured at a series of scan rates, including 0.05, 0.1, 0.15, 0.2, and 0.25 mV s^{-1}, as shown in Figure 3e,f. The Na-ion diffusion coefficient is investigated by the classical Randles–Sevcik equation (Equation 1).

$$I_p = 2.69 \times 10^5 n^{3/2} A {D_{Na}}^{1/2} \upsilon^{1/2} C_{Na} \tag{1}$$

here, I_p represents the peak current (mA), n is the charge transfer coefficient, A denotes the electrode surface area, D_{Na} is the diffusion coefficient of Na-ion, υ is the scan rate (mV/s), and C_{Na} is the concentration of Na-ion. Consequently, the diffusion coefficient D_{Na} for the NFM and NFMAB cathodes can be determined by linearly fitting the relationship between I_p versus $\upsilon^{1/2}$. In Figure 3g, the values of $I_p/\upsilon^{1/2}$ for the anodic and cathodic of NFMAB cathode are 1.29×10^4 and 8.16×10^3, respectively, much higher than those of the NFM cathode (5.98×10^3 and 4.30×10^3). It clearly indicates that the NFMAB cathode delivers faster Na-ion diffusion than NFM cathode. This enhancement can be ascribed to the expansion of Na-ion channels resulting from B-element doping, which significantly boosts the kinetic performance.

Moreover, the Na-ion diffusion coefficient (D_{Na}) can be calculated by the galvanostatic intermittent titration technique (GITT) (Figure 3h,i, Equation 2).

$$D_{Na} = \frac{4}{\pi\tau}\left(\frac{m_B V_m}{M_B A}\right)^2 \left(\frac{\Delta E_s}{\Delta E_\tau}\right)^2 \tag{2}$$

In this context, D_{Na} represents the diffusion coefficient of Na-ion, τ denotes the titration time, m_B is the mass of the active material, V_m and M_B correspond to the molar volume and molar mass, respectively. Additionally, A signifies the electrode surface area, ΔE_S is the quasi-equilibrium potential difference before and after the current pulse, and ΔE_τ represents the potential difference during the current pulse. As shown in Figure 3i, the D_{Na} of NFMAB cathode is in the range of 10^{-10}–10^{-8} cm^2 s^{-1} in the sodiation and desodiation processes. During the whole charging/discharging process, the NFMAB cathode offers a higher D_{Na} compared with NFM cathode.

Concurrently, the variable-temperature EIS of the NFM and NFMAB samples were measured over a range of 298.15–318.15 K, as shown in Figure 3j,k. Equivalent circuit models are presented in Figures S6 and S7 (Supporting Information) illustrates the Nyquist fitting process. Detailed fitting parameters are summarized in Table S4 (Supporting Information).

The apparent *Ea* of the battery can be determined by fitting the variable-temperature EIS results by the Arrhenius equation. The *Ea* values can be calculated based on Equations 3 and 4:

$$k = A \times exp(\frac{-Ea}{RT}) \tag{3}$$

$$ln\,(k) = ln\,(A) - \frac{Ea}{RT} \tag{4}$$

k is the rate constant of the reaction, R is the molar gas constant, T is the thermodynamic temperature, and A is the pre-exponential factor. The calculated *Ea* for the charge transfer process in the NFMAB battery is 69.2 kJ mol^{-1}, significantly lower than that of the reference sample (90.5 kJ mol^{-1}). The low *Ea* suggests that the material exhibits highly efficient interfacial kinetics and reduced sensitivity to temperature variations. These characteristics are highly desirable for batteries intended for high-power and wide-temperature-range applications.

To gain insight into the structural evolution during cycling and the origins of electrochemical performance, in situ XRD patterns of NFM and NFMAB half cells were collected at 0.1C within the voltage range of 2.0–4.3 V (**Figure** 4a–d). Upon the initial charging, the (003) and (006) characteristic peaks of NFM and NFMAB electrodes shift to low angles, while the (104) peak shifts to high angles. The structural evolution during charging is governed by competing lattice responses, as the following discussion. Na-ion extraction induces enhanced electrostatic repulsion between adjacent oxygen layers, driving *c*-axis expansion, while concurrent Ni^{2+} oxidation initiates ab plane contraction through metal-oxygen bond shortening.[28] As charging proceeded, the (104) peak characteristic of the O3 crystal structure vanished, marking the formation of a new hexagonal P3 phase via TMO_6 slabs sliding. The coexistence of the O3 and P3 phases at a low voltage (≈3.2 V) was observed in the NFM sample, in which the coexistence of two phases may induce structural stress at grain boundaries. Conversely, the NFMAB electrode exhibits a more rapid phase transformation, indicating that the mitigation of such phase coexistence can reduce structural stress and crack formation, thereby enhancing cycling stability. Meanwhile, the rapid phase transformation can alleviate the obstruction of Na-ion transport at grain boundaries, improving high-Na-ion rate capability. When the charging voltage exceeds 4.0 V, the (003) diffraction peak undergoes a shift toward high angles, signaling the emergence of an OP2 phase accompanied by rapid

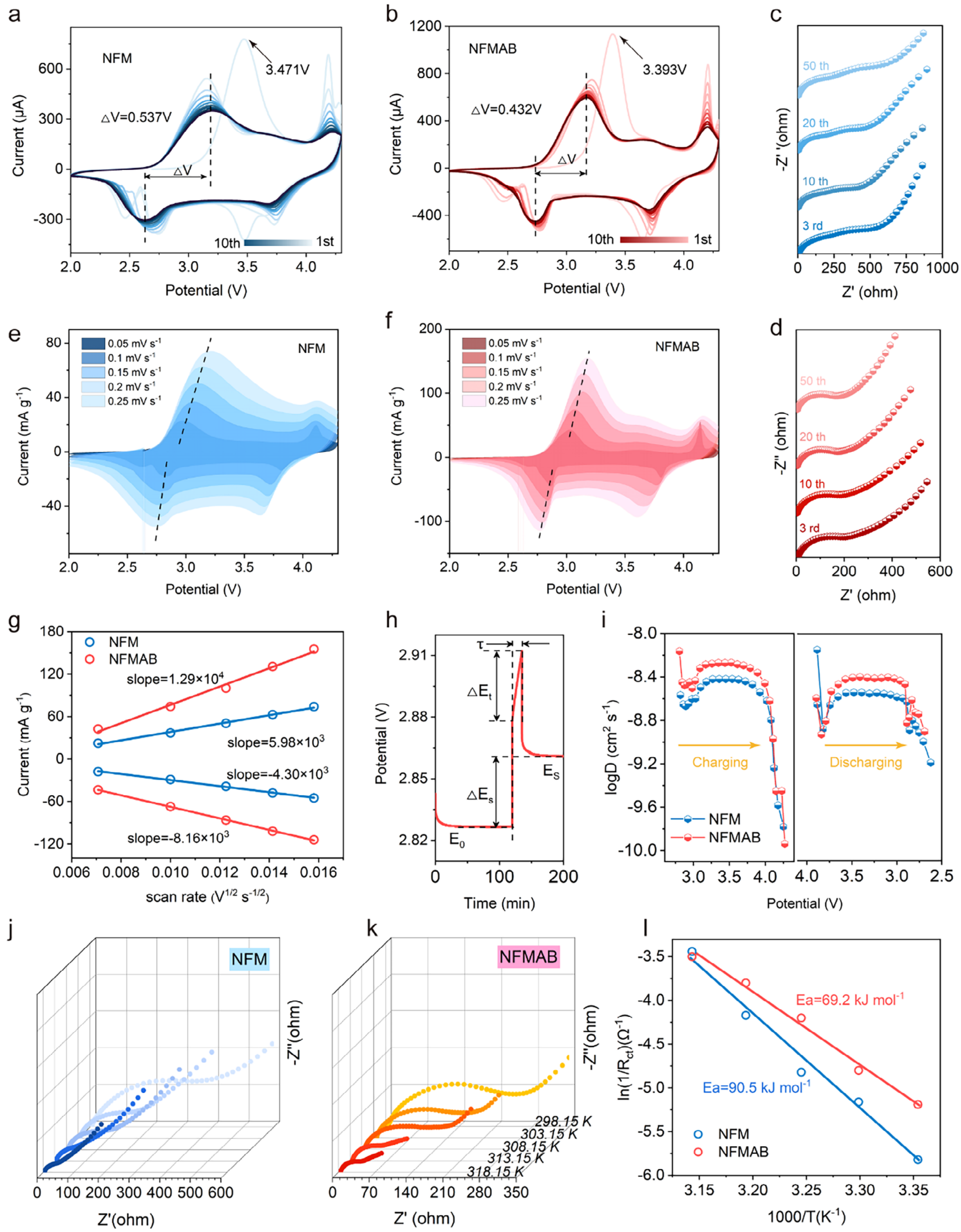

Figure 3. a) CV profiles of NFM and b) NFMAB cathodes at a scan rate of 0.1 mV s^{-1}; c) EIS of NFM and d) NFMAB cathodes at different cycling stages; Variable scan-rate CV tests for e) NFM (0.05–0.25 mV s^{-1}) and f) NFMAB (0.05–0.25 mV s^{-1}); g) Anodic and cathodic peak current ratios ($Ip/\upsilon^{1/2}$) for NFM and NFMAB cathodes derived from CV analysis; h) Potential response profiles of NFM and NFMAB cathodes during GITT measurements; i) Na-ion diffusion coefficients as a function of voltage calculated via GITT; Temperature-dependent EIS of j) NFM (298.15–318.15 K) and k) NFMAB (298.15–318.15 K); l) Corresponding Arrhenius curves and comparison of activation energies of NFM and NFMAB cathodes.

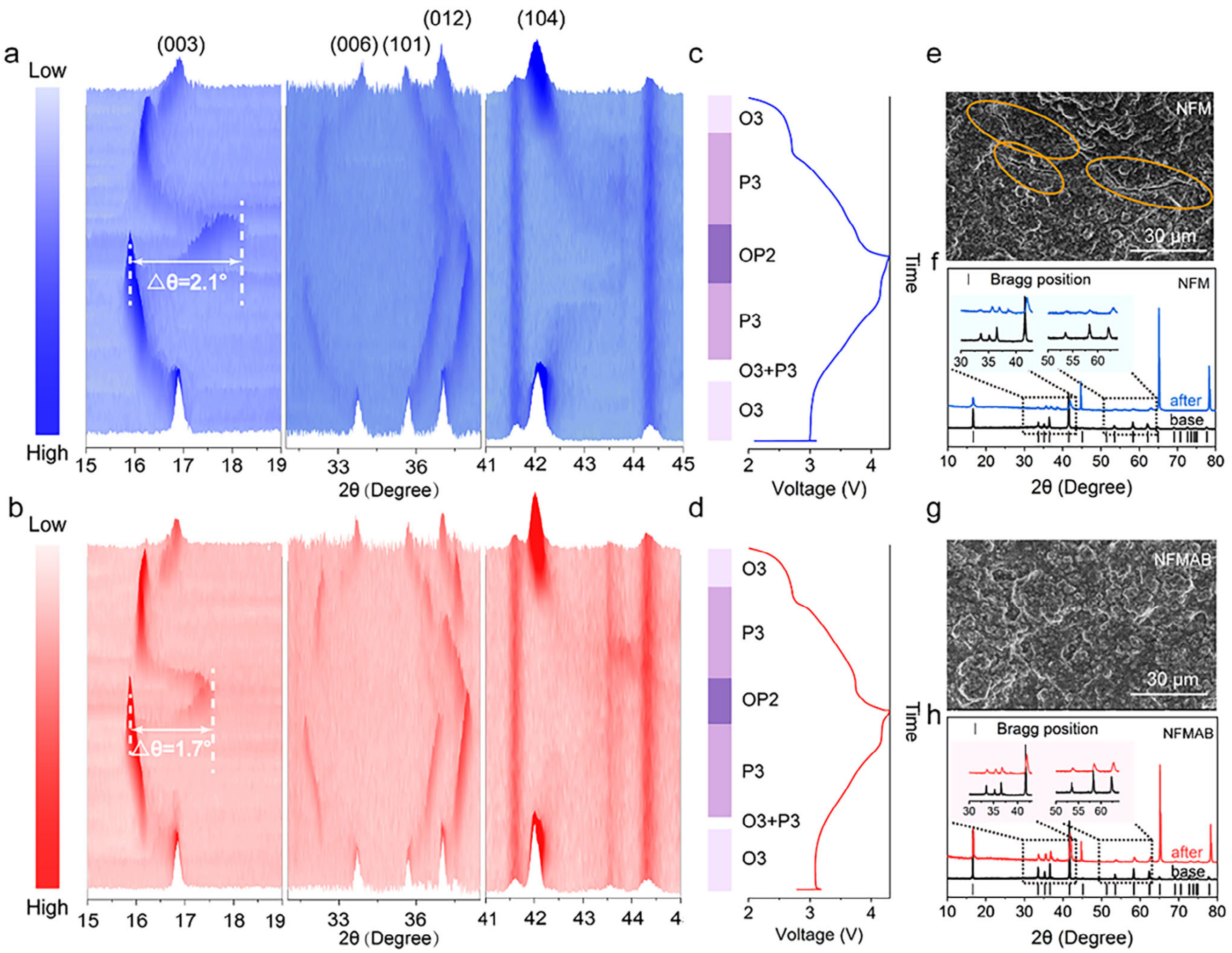

Figure 4. In situ XRD pattern during the first cycle of the a) NFM and b) NFMAB half cells under a current rate of 0.3C at voltage range between 2 and 4.3 V; Charge–discharge curves of c) NFM and d) NFMAB half cells for in situ XRD measurements; e) SEM and f) XRD characterization of NFM electrodes after 200 cycles; g) SEM and h) XRD characterization of NFMAB electrodes after 200 cycles.

c-axis contraction. The pronounced lattice distortion detrimentally compromises structural integrity, thus amplifying lattice strain and propagating particle cracking. Notably, the maximum angular displacement ($\Delta\theta = 1.7°$) of the (003) peak in NFMAB electrodes starkly contrasts with that of pristine NFM ($\Delta\theta = 2.1°$), demonstrating enhanced structural coherence in the co-doped system. Maximum angular shifts of various characteristic peaks are provided in Table S5 (Supporting Information). During discharge, the OP2 phase undergoes sequential transformation to a P3 phase, followed by O3 phase reconstitution below 2.4 V. This establishes a reversible O3↔P3↔OP2 phase evolution pathway during cycling (O3→O3+P3→P3→OP2→P3→O3). Crucially, Al doping reinforces TM—O bonding, leading to the suppressed *c*-axis contraction at high voltages (>4.0 V) and the minimized metastable two-phase coexistence- key factors for enhancing structural stability. Al doping within the TM layers enhances structural stability by mitigating interlayer slippage in TMO_6 slabs during charge/discharge cycling, a mechanism attributed to the stronger Al—O bonding energy (512 kJ mol^{-1}) compared with Ni—O (392 kJ mol^{-1}), Fe—O (409 kJ mol^{-1}), and Mn—O (402 kJ mol^{-1}). This atomic-scale stabilization mechanism directly correlates with the superior long-term cyclability of Al-doped systems demonstrated in Figure 2a.

To further investigate the structural stability of the battery after cycling, NFM and NFMAB half-cells, which had undergone 200 cycles at 2C, were analyzed using SEM and ex situ XRD (Figure 4e–h). As demonstrated in Figure 4e,g, the NFMAB electrode exhibits significantly mitigated cracks after cycling compared with the pristine NFM sample. It indicates that strengthened TM—O bonds suppress slab sliding and mitigate volume fluctuations during cycling, preventing crack formation from particle-electrolyte contact failure due to volumetric changes. Additional SEM images of other selected areas are provided in Figure S8 (Supporting Information) to corroborate the structural stability of the NFMAB electrode after cycling. The XRD patterns shown in Figure 4f,h reveal that the NFMAB electrode retains integrally the O3 phase after 200 cycles, as evidenced by the characteristic peaks of the O3-type layered oxide between 30° and

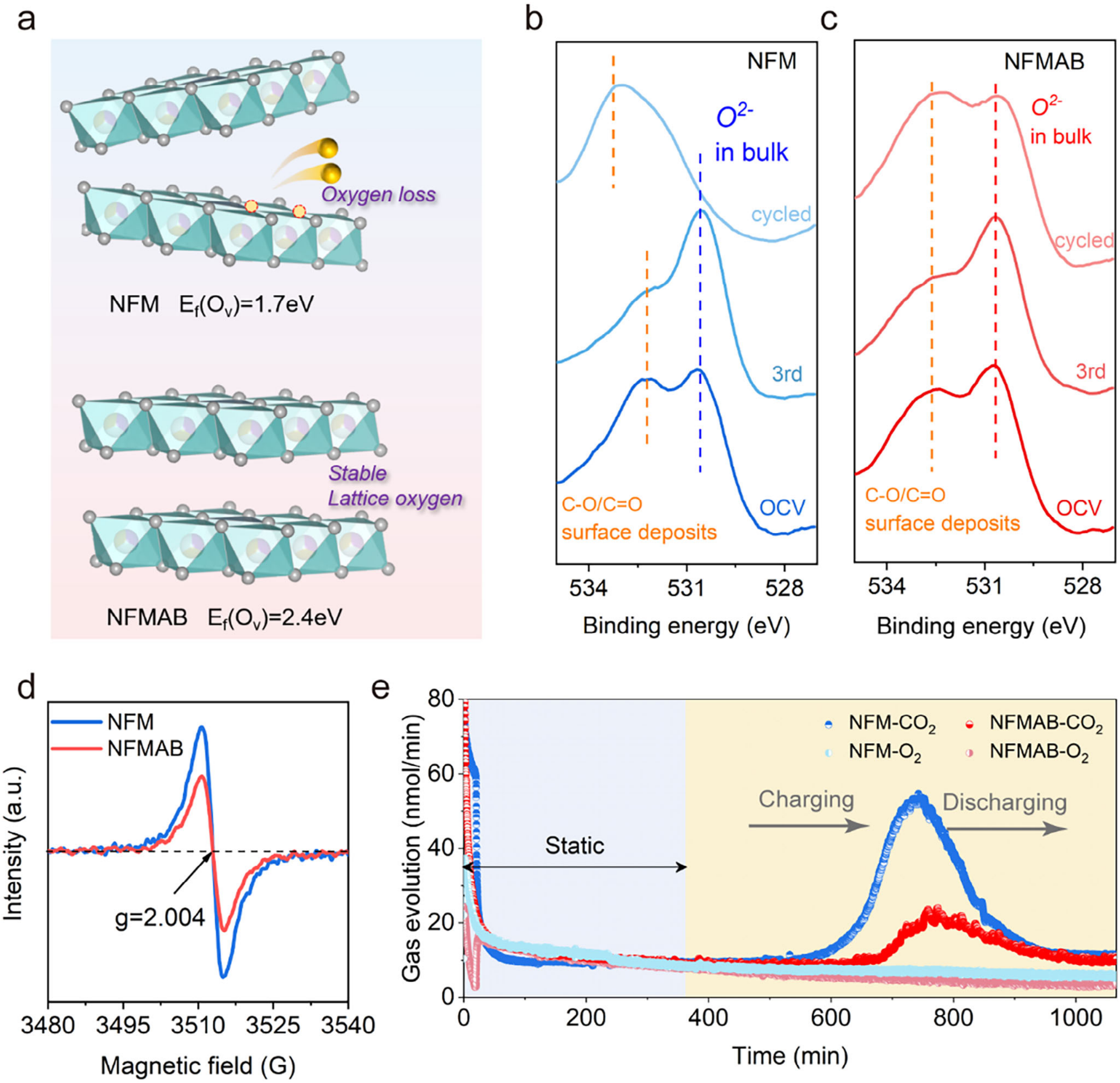

Figure 5. a) Oxygen vacancy formation energies of NFM and NFMAB materials; b) O 1s XPS spectra of the NFM and c) NFMAB electrodes at pristine state, after 3 cycles, and after 200 cycles; d) EPR spectra of NFM and NFMAB electrodes after 100 cycles; e) Differential electrochemical mass spectrometry profiles during the first charge/discharge cycle under a current rate of 0.1C at a voltage range between 2 and 4.3 V.

45° and between 50° and 65°, demonstrating significant structural reversibility. In contrast, the NFM displays progressive peak displacement and intensity attenuation, indicating irreversible phase degradation. Comparative analysis of single doping cathodes (Figure S9a,b, Supporting Information) further demonstrates the pivotal role of Al: NFMA retains structural coherence comparable to NFMAB after cycling, while NFMB suffers accelerated collapse, highlighting Al doping as the dominant stabilizing factor. These findings establish that Al incorporation fundamentally stabilizes the TMO_6 slab architecture through strong orbital hybridization, effectively relieving lattice stress from electrochemical cycling. This atomic-scale reinforcement plays a pivotal role in preserving the structural coherence and mechanical integrity of NFMAB, underpinning its exceptional long-term cycle.

Layered oxide cathodes, such as NFM, are prone to lattice oxygen activation during cycling at high voltages (>4.1 V), leading to the release of gases like O_2 and CO_2. We explored the properties of NFM and NFMAB cathodes via multi-scale characterization and revealed the mechanism of doping enhanced high-voltage oxygen stability. DFT calculations (**Figure 5a**) show that the oxygen vacancy formation energy in NFMAB (2.4 eV) is much higher

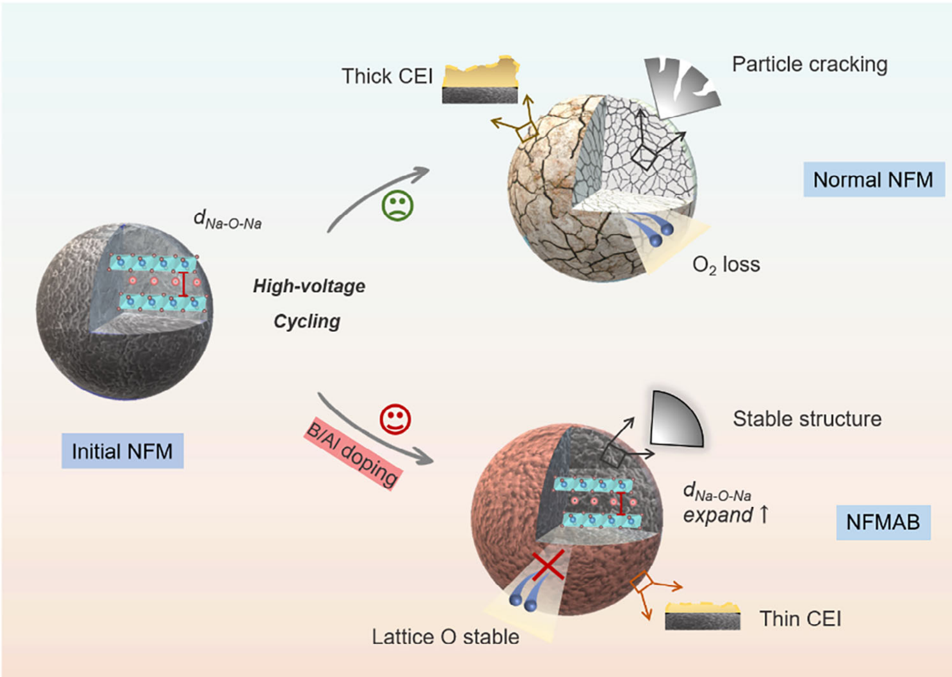

Figure 6. Schematic Illustration of stabilizing phase transition and lattice oxygen in B and Al co-doping materials.

than that in pristine NFM (1.7 eV). From a thermodynamic perspective, B and Al doping can effectively increase the formation barrier of oxygen vacancy and suppress the tendency of lattice-oxygen escape. We performed Ar^+ ion sputtering-assisted X-ray photoelectron spectroscopy (XPS) on NFM and NFMAB samples after various cycles (Figure 5b,c). The peak at 532.1 eV in the O 1s spectrum is attributed to surface decomposition products of the electrolyte, while the peak at 529.5 eV is due to lattice oxygen (O^{2-}) in the bulk of the layered oxide.[29] Both pristine and activated NFM and NFMAB samples show distinct O^{2-} peaks. However, after cycling, the O^{2-} peak in NFM decreases significantly, suggesting the oxidation of O^{2-} to a high valence state. While the O^{2-} peak in NFMAB remains relatively unchanged, demonstrating that the strong B—O bond stabilizes lattice oxygen and inhibits its irreversible oxidation.

As shown in Figure 5d, both NFM and NFMAB cathodes exhibit clear electron spin resonance signals ($g = 2.004$), attributed to electrons trapped in oxygen vacancies.[30] The weak electron paramagnetic resonance (EPR) signal in NFMAB indicates a low oxygen vacancy concentration, suggesting reduced active oxygen escape at high voltages, consistent with the XPS results. Differential electrochemical mass spectrometry (DEMS) was used to measure gas generation during cycling (Figure 5e). The charge–discharge curves in the DEMS test are shown in Figure S10 (Supporting Information). During the first charge–discharge, CO_2 is detected in both NFM and NFMAB, but O_2 is not detected. It is known that the lattice oxygen in the NFM-based layered oxide will undergo an oxidation reaction under high voltage.[9] At high voltage, due to the chemical reaction between reactive lattice oxygen and electrolyte, the oxygen species may be converted into CO_2 or CO.[31] In the first cycle of NFM half cells at 0.1C between 2 and 4.3 V, significant CO_2 release (peak area 5332 nmol) is detected, corresponding to irreversible oxygen loss and electrolyte decomposition. In NFMAB half-cells, CO_2 release was reduced to 1540 nmol (28.8% of the control), proving that doping effectively suppresses gas generation.

This schematic elucidates the synergistic B and Al co-doping strategy that optimizes the electrochemical performance of O3-type layered NFM oxide cathodes under high-voltage operation. The pristine NFM suffers from sluggish Na-ion diffusion kinetics, particle fracturing caused by phase-transition-induced stresses, irreversible lattice oxygen loss, and thick CEI formation during high-voltage cycling. The co-doping strategy enhances Na-ion diffusion through expanded Na—O interlayer spacing, while Al^{3+} occupation in TM layers reinforces Al—O bonds to inhibit interlayer slippage and phase transitions. Concurrently, B^{3+} establishes robust B—O covalent bonds while increasing oxygen vacancy formation energy to 2.4 eV, effectively mitigating lattice oxygen depletion and electrolyte decomposition. These synergistic modifications stabilize the bulk structure and improve interfacial compatibility under high-voltage operation, demonstrating an atomic-scale bond-energy engineering paradigm for developing high-energy-density sodium-ion cathodes (**Figure 6**).

3. Conclusion

In conclusion, this work demonstrates that the atomic-scale bond-energy engineering via B and Al co-doping fundamentally resolves the intrinsic limitations of O3 layered oxides under high-voltage operation. By strategically positioning Al^{3+} in TM layers and B^{3+} at TM—O interstices, we achieve dual stabilization of bulk structure and interfacial chemistry-Al mitigates phase-transition-induced strain through reinforced TM—O bonding, while B suppresses lattice oxygen activation via strong covalent interactions. The ICE of the NFMAB cathode is significantly improved, demonstrating a substantial reduce of capacity loss which is attributed to the irreversible lattice oxygen release and the electrolyte decomposition catalyzed by reactive oxygen species. After 200 cycles, the NFMAB electrode retains 63.1% of its initial capacity, markedly outperforming the pristine NFM (31.3%), highlighting its enhanced structural stability during cycling. Furthermore, the NFMAB cathode delivers a reversible discharge capacity 2.6 times higher than that of the pristine NFM at 5C. Even at an ultrahigh current density of 10C, it maintains a high capacity of 87.6 mAh g^{-1}, confirming its exceptional rate capability. In summary, this work ingeniously devises an efficient strategy to collaboratively address bulk and interfacial issues in O3-layered oxides at high voltage via B and Al dual-elements doping. This rational doping strategy offers new insights for realizing high energy density in sodium ion batteries and further accelerating their commercialization.

4. Experimental Section

Material Synthesis: The O3-$NaNi_{1/3}Fe_{1/3}Mn_{1/3}O_2$ cathode material was fabricated via a straightforward wet ball milling combined with a high-temperature solid phase approach. To begin with, specific quantities of Na_2CO_3 and $Ni_{1/3}Fe_{1/3}Mn_{1/3}(OH)_2$ were precisely weighed out, with Na_2CO_3 being in 5% excess according to the molar ratio. Subsequently, the resultant mixture was initially ground using a mortar and then transferred into an agate ball mill jar. Acetone was introduced as a dispersant, following which the mixture was subjected to ball milling with zirconium for a duration of 6 h. Thereafter, the milled mixture was calcined in air within a muffle furnace at a heating rate of 5 °C per minute. It was heated up to 900 °C and maintained at this temperature for 15 h. Subsequently, upon its natural cooling down to 200 °C, it was transferred into an argon-filled glove box. The obtained O3-$NaNi_{1/3}Fe_{1/3}Mn_{1/3}O_2$ cathode material was expressed in NFM. O3-$Na(Ni_{1/3}Fe_{1/3}Mn_{1/3})_{0.98}Al_{0.02}B_{0.02}O_2$ cathode electrodes were additionally fabricated through the identical synthetic approach. During the preparation procedure, Na_2CO_3, $Ni_{1/3}Fe_{1/3}Mn_{1/3}(OH)_2$, Al_2O_3, and B_2O_3 were added in accordance with the molar ratio (with Na_2CO_3 being in 5% excess). The obtained O3-$Na(Ni_{1/3}Fe_{1/3}Mn_{1/3})_{0.98}Al_{0.02}B_{0.02}O_2$ cathode material was expressed in NFMAB. The O3-$Na(Ni_{1/3}Fe_{1/3}Mn_{1/3})_{0.98}Al_{0.02}O_2$ and O3-$NaNi_{1/3}Fe_{1/3}Mn_{1/3}B_{0.02}O_2$ cathodes were also prepared in the same way and were abbreviated as NFMA and NFMB.

Materials Characterization: The successful preparation of different cathode materials was characterized by X-ray diffraction (XRD, Smartlab). Ex situ XRD tests were conducted on the materials, where the materials were subjected to constant-current charging at different cut-off voltages and maintained under constant voltage, aiming to analyze the structural changes of the materials during the charging and discharging processes. The structural characteristics of the prepared materials after cycling were investigated by scanning electron microscopy (SEM, Apreo S LoVac). Transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) observations on the prepared materials were carried out using JEM-F200 at an accelerating voltage of 200 kV. Moreover, the elemental distribution of the materials was characterized by an energy dispersive spectrometer (EDS). Electron paramagnetic resonance (EPR) was conducted on a Bruker EMX PLUS spectrometer. The alterations in oxygen within the bulk lattice were quantified by X-ray photoelectron spectroscopy (XPS, Thermo-Fisher Scientific) employing an Al Kα radiation source. Depth profiles were then acquired with the assistance of Ar^+ etching. The energy of the XPS data was calibrated in accordance with the C 1s spectra at 284.8 eV. Prior to the XPS measurement, the cycled samples were rinsed three times with propylene carbonate (PC) to eliminate the residual electrolyte salt. Differential electrochemical mass spectrometry (DEMS) testing was carried out using the QAS 100 Li equipment from Shanghai LingLu Company.

Computational Details: The structure optimization was conducted with materials studio 8.0 using the CASTEP package. Generalized gradient approximation with Perdew–Burke–Ernzerhof functional (GGA-PBE) was used for the exchange-correlation energy.

The doping energy was defined by the following formula:

When Al was doped into the NFM cathode, considering its relatively large radius, its substitution for TM sites or Na sites was mainly considered.

$$\begin{aligned} &\Delta E\,(Al_1) = E\,(NFMA) - [E\,(NFM) - \mu(TM)] - \mu(Al) = \\ &E\,(NFMA) - E\,(NFM) + \mu(TM) - \mu(Al) \\ &\Delta E\,(Al_2) = E\,(AlNFM) - [E\,(NFM) - \mu(Na)] - \mu(Al) = \\ &E\,(AlNFM) - E\,(NFM) + \mu(Na) - \mu(Al) \end{aligned} \tag{5}$$

The doping energies of $\Delta E(Al_1)$ and $\Delta E(Al_2)$ correspond to the energies associated with Al atom doping at sites TM and Na, respectively. E(NFM), E(NFMA), and E(AlNFM) are the energies of the initial, Al replaced TM site, Al replaced Na site, respectively. $\mu(i)$ is the atomic chemical potential of the species i, including the TM atom, Na atom, and Al atom.

Owing to the smaller ionic radius of B, it is considered that B may substitute for the Na site or the TM site, or be doped within the Na—O layer or the TM—O layer.

$$\begin{aligned} &\Delta E\,(B_1) = E\,(NFMB) - [E\,(NFM) - \mu(TM)] - \mu(B) = \\ &E\,(NFMA) - E\,(NFM) + \mu(TM) - \mu(B) \\ &\Delta E\,(B_2) = E\,(BNFM) - [E\,(NFM) - \mu(Na)] - \mu(B) = \\ &E\,(AlNFM) - E\,(NFM) + \mu(Na) - \mu(B) \\ &\Delta E\,(B_3) = E\,(NFMB_{TM-O}) - E\,(NFM) - \mu(B) \\ &\Delta E\,(B_4) = E\,(B_{Na-O}NFM) - E\,(NFM) - \mu(B) \end{aligned} \tag{6}$$

The energies $\Delta E(B_1)$, $\Delta E(B_2)$, $\Delta E(B_3)$, and $\Delta E(B_4)$ correspond to the doping energies of B atoms substituting for the Na site, the TM site, or being doped within the Na—O layer or the TM—O layer, respectively.

O_2 release energy was given by the equation as follows:

$$\begin{aligned} &\Delta E\,(NFM_{-O}) = E\,(NFM - O) + 1/2E\,(O_2) - E\,(NFM) \\ &\Delta E\,(NFMAB_{-O}) = E\,(NFMAB - O) + 1/2E\,(O_2) - E\,(NFMAB) \end{aligned} \tag{7}$$

where E(NFM-O) and E(NFMAB-O) represent the energies of the fully relaxed oxygen-deficient structures following the removal of the oxidized oxygen from the materials; E(NFM) and E(NFMAB) correspond to the total energies of the fully relaxed NFM and NFMAB structures, respectively, each incorporating an oxygen vacancy; $E(O_2)$ represents the chemical potential of oxygen at room temperature; $\Delta E(NFM_{-O})$ and $\Delta E(NFMAB_{-O})$ denote the oxygen vacancy formation energies for the NFM and NFMAB cathodes, respectively.

Electrochemical Characterization: The electrochemical performance was evaluated with CR2032 (Canrd Technology Co., Ltd.) coin cells. The cathode electrode was composed of cathode material powder, Super P, and polyvinylidene fluoride (PVDF) in the weight ratio of 8:1:1, where a homogeneous slurry was prepared with 1-methyl-2-pyrrolidone (NMP) as the solvent. Then, the slurry was evenly coated on the aluminum (Al) foil. After drying in a vacuum oven at 70 °C, the material was cut into electrodes with a diameter of 12 mm, and the mass loading was ≈3.8 mg cm^{-2}. The coin cell was assembled in a glove box filled with argon gas (O_2 and water < 1 ppm). 1 M $NaClO_4$ in PC+5% FEC was used as the electrolyte, and glass fiber (Whatman, GF/D) as the separator. Galvanostatic charge/ discharge measurements were carried out at room temperature on a NEWARE test system. The half cell was cycled at 0.1C (1C = 130 mA g^{-1}) for 3 cycles and 2C for the following cycles within the potential range of 2.0–4.3 V. Ivium electrochemical workstation was utilized to characterize the CV curves (voltage window: 2–4.3 V). EIS was performed at the CHI660 electrochemical workstation in the frequency range of 10^5–0.01 Hz.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

aluminum, bond-energy engineering, boron, high-voltage, sodium-ion battery

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